

Team Synapz — Autonomous Construction Safety Intelligence Platform

Startathon Master Research, Product, Technical & Business Readiness Document

Research baseline: 4 September 2026

Team: Team Synapz — 3 members

Event constraint: 30-hour build after kickoff

Product thesis: Turn existing construction-site CCTV from passive evidence into an active safety intelligence layer.

0. Executive decision

The product we should build

A distributed, edge-first construction safety intelligence platform that observes existing CCTV, converts visual detections into structured safety events, uses specialized agents to correlate and prioritize risk, delivers immediate local warnings through site kiosks, and escalates actionable incidents to supervisors with concise mobile alerts and evidence clips.

The core loop is:

Observe → Understand → Reason → Act → Verify → Learn

The MVP should not attempt to automate every possible construction-safety task. It should prove one compelling proposition:

A site can move from “someone may eventually notice the CCTV” to “the system detects a high-risk event, understands its context, warns the people nearby, and records evidence automatically.”

Critical strategic conclusion from competitor research

The basic category already exists. viAct, Protex AI, Intenseye, Xorth and WorkVis all demonstrate substantial overlap with CCTV-based PPE, zone, vehicle, fall and hazard monitoring. Therefore **“AI detects construction hazards” is not our moat.**

Our differentiated product thesis should be:

- **Construction-first site intelligence**, not a generic workplace CV product.
- **Worker-facing local intervention** through physical kiosks / speakers, not only a remote dashboard.
- **Distributed local safety nodes** that can act near the risk while a site-level coordination layer correlates events.
- **Edge/offline-first operation** for sites with poor connectivity and privacy requirements.
- **Low-friction retrofit**: connect to CCTV/IP cameras already installed instead of requiring a new camera network.
- **Site-specific safety graph**: zones, camera coverage, rules, events, interactions and historical hotspots become a persistent model of the site.
- **India-first commercial wedge**: practical deployment, localized alerts, affordable site-level pricing and support for existing CCTV/NVR infrastructure.

None of these should be presented as a claim that competitors cannot do them. They are the **product choices we intend to make differently** and the hypotheses we must validate.

1. Startathon strategy

The judging rubric emphasizes six dimensions:

Dimension	Weight	What Team Synapz must prove
Problem insight	20%	We understand the actual site workflow, users and alternatives
Validation & learning	20%	We tested important assumptions and can explain what changed
Solution & value proposition	15%	The core value is obvious and scope is disciplined
Product & technical execution	25%	The end-to-end workflow works reliably
Team execution & ownership	10%	Three people make deliberate decisions and respond to evidence
Communication & continuation	10%	Clear story, honest limitations and credible next experiment

Implication: We should not spend 30 hours adding 20 flashy detections. A measured, working 3–4 scenario safety loop can score better than a huge but fragile architecture.

2. Problem insight

2.1 The current safety loop

A simplified construction-site safety workflow is:

- Project owner / main contractor establishes site safety requirements.
- Safety/EHS personnel and supervisors identify hazards and define controls.
- Contractors/subcontractors brief workers and coordinate work.
- Workers perform tasks while supervisors and safety personnel conduct inspections, observations and toolbox talks.
- CCTV may record site activity, but human operators cannot continuously interpret every feed with equal attention.
- When a hazard is noticed, someone must determine severity and communicate it.
- A supervisor or worker takes corrective action.
- Serious incidents and near misses are documented and investigated.
- Patterns are reviewed to improve controls and training.

OSHA's construction guidance explicitly emphasizes management commitment, worker participation, hazard identification, prevention/control, emergency planning, inspections and incident investigation. It also emphasizes coordination among employers/contractors and participation by contractors and subcontractors. [Source: OSHA Construction Safety and Health Program — <https://www.osha.gov/etools/construction/safety-health-program>]

In India, the BOCW Act provides for safety committees and safety officers in establishments with 500 or more building workers, and establishes accident-notification obligations for qualifying incidents. The Central Rules also specify safety-officer and

accident-reporting requirements. [Source: OSHA Construction Safety and Health Program — <https://www.osha.gov/etools/construction/safety-health-program>]

2.2 Where our product fits

We are **not replacing the safety program**.

We are inserting an always-on intelligence layer between:

site activity → hazard recognition → immediate local intervention → supervisor escalation → evidence → learning

That means the product augments:

- Safety officers
- Site supervisors
- Control-room staff
- Contractors
- Workers

It should not claim to replace human accountability or statutory safety responsibilities.

3. Detailed competitor teardown

3.1 viAct

What viAct does

viAct markets AI video analytics for workplace/construction safety, including:

- PPE detection
- Danger-zone alerts
- Work-at-height monitoring
- Fleet management
- Confined-space monitoring
- Environmental monitoring
- Real-time alerts
- Site dashboards
- Existing IP camera/CCTV integration

Its area-control product includes geofencing, unauthorized access, danger-zone intrusion and vehicle-human proximity alerts. viAct also describes a workflow connecting IP cameras/CCTV to its AI cloud and a client monitoring layer.

Strengths

- Strong construction positioning.
- Broad detection catalogue.

- Real-time alerts.
- Existing-camera integration.
- Dashboard and documentation.
- Mature commercial positioning.

Weakness / opening for us

We should **not** claim viAct is only a dashboard or that it cannot alert locally. It clearly supports real-time intervention. Instead, our opening is a more explicit **distributed safety-node architecture**, local kiosk UX, site-level event graph and India-first deployment model.

What we learn

A serious construction AI product needs more than PPE detection. It needs zones, vehicles, work-at-height, evidence and workflow.

3.2 Protex AI

What Protex does

Protex has moved beyond individual CV detections into “site intelligence.” It connects existing cameras with enterprise data such as EHS, WMS, MES, TMS and CMMS systems; detects safety and operational risks; performs pattern/root-cause analysis; supports configurable rules; offers heatmaps and reporting; and markets custom agentic workflows.

Its vehicle-control system handles vehicle/pedestrian proximity, restricted areas and configurable zone rules. Its platform states that detection-to-signal can take 1–2 seconds for real-time use cases.

Strengths

- Very strong site-intelligence concept.
- Existing-camera integration.
- Edge/on-site processing.
- EHS integrations.
- Heatmaps/trends.
- Agentic workflows.
- Mature enterprise privacy story.

Weakness / opening for us

Protex is a **major benchmark competitor**, so we should never say “agents are our unique feature.” Protex already markets agentic workflows and site intelligence.

Our differentiation hypothesis is **construction-specific distributed safety nodes + physical local intervention + retrofit economics + India-first deployment**.

What we learn

The industry is moving from “detect” to “detect → understand → act → learn.” Our architecture should do the same.

3.3 Intenseye

What it does

Intenseye positions itself as an EHS platform with computer vision models covering 50+ leading indicators. It monitors:

- Area controls
- Vehicle controls
- Behavioral safety
- PPE
- Ergonomic risks
- Safety conditions
- Video evidence
- Analytics/reporting

It emphasizes 24/7 visibility, EHS data and enterprise safety workflows.

Strengths

- Large detection breadth.
- Strong EHS orientation.
- Analytics and reporting.
- Enterprise credibility.
- Privacy-conscious positioning.

Weakness / opening

Again, not “they don’t detect hazards.” They clearly do. Our wedge is the physical, localized safety-intervention network and construction-site-specific operating model.

What we learn

The buyer is not buying object detection. They are buying **risk reduction and actionable EHS intelligence**.

3.4 Xorth

What it does

Xorth is particularly important because it is **India-focused** and directly overlaps with our idea.

It markets existing-CCTV AI for Indian construction sites and describes 21 detection models across safety, security and operations. Its construction page includes:

- PPE compliance
- Harness/work-at-height
- Danger-zone breaches
- Fire/smoke
- Perimeter breach
- Unauthorized entry
- Vehicle plate recognition
- Material theft

- Housekeeping/waste
- Zone occupancy
- Vehicle/load-related monitoring

Xorth also describes edge-native/on-prem/private-cloud deployment and says inference can run on a GPU server inside the site's network.

Strengths

- India-specific construction positioning.
- Existing CCTV/NVR.
- Large construction model catalogue.
- On-prem/edge story.
- Compliance/evidence angle.
- Safety + security + operations.

Weakness / opening

Xorth is our strongest India-market warning. We cannot claim “Indian construction sites don't have this.”

Our wedge must therefore be narrower:

- Worker-facing kiosk intervention
- Distributed site nodes
- Explicit event-to-action orchestration
- Site safety graph
- Offline-first local operation
- Affordable, transparent deployment
- Potentially multilingual voice warnings
- Stronger focus on interaction risk, not only isolated detections

What we learn

Local market knowledge and deployment constraints can be as important as model accuracy.

3.5 WorkVis

What it does

WorkVis uses existing/new surveillance camera feeds to identify hazards and alert workers and safety managers. It covers:

- PPE
- Vehicle collision
- Crane safety
- Site inspection
- Hot work
- Fall detection
- Machine guarding

- Confined-space entry
- Housekeeping

A particularly relevant feature: WorkVis explicitly describes **local audio/visual worker alerts**, followed by supervisor notification if corrective action is not taken.

Strengths

- Very strong worker-facing intervention model.
- Existing camera feeds.
- Local alerts.
- Escalation chain.
- Reporting/analytics.
- Industrial safety experience.

Weakness / opening

This eliminates “local alerts” as a unique claim.

Our distinction must instead be the **construction-specific multi-camera/site intelligence architecture**, distributed nodes, event graph and India-first deployment/economics.

What we learn

A good safety system should warn the person near the hazard first, then escalate if the hazard persists.

4. Competitive positioning

Capability	viAct	Protex	Intenseye	Xorth	WorkVis	Team Synapz hypothesis
Existing CCTV	✓	✓	✓	✓	✓	✓
PPE	✓	✓	✓	✓	✓	✓
Zone control	✓	✓	✓	✓	✓	✓
Vehicle/pedestrian	✓	✓	✓	✓	✓	Core
Fall/work-at-height	✓	✓/config.	✓	✓	✓	Selective MVP
Local worker alert	✓/workflow	✓/workflow	✓	✓	✓	Core UX
EHS/reporting	✓	✓	✓	✓	✓	MVP-lite
Heatmaps	✓	✓	✓	✓	✓	Core
Agentic workflows	Not primary positioning	✓	AI analytics	AI platform	Workflow escalation	Purposeful agents
Distributed swarm framing	Not core	Agentic/site intelligence	Not core	Edge platform	Not core	Core architecture thesis
Construction-first	✓	Broad industrial	Broad EHS	✓	Industrial	Core
India-first	Not core	Not core	Not core	✓	Not core	Core wedge
Physical kiosk	Not the core differentiation	Not core	Not core	Not core	Local speaker/alert	Core product interface

Conclusion: our moat is not “YOLO + agents.” It must become the combination of **site-specific data, deployment know-how, safety-event graph, local intervention, customer workflows and accumulated site history**.

5. Feature prioritization matrix

Score: **Impact (1–5) × Demo feasibility (1–5) × Evidence availability (1–5)**, with a qualitative priority.

Feature	Impact	Feasibility	Evidence	Priority	MVP?
Person detection/tracking	5	5	5	P0	YES
Helmet / PPE	5	5	5	P0	YES
Vehicle detection	5	5	5	P0	YES
Vehicle–person proximity	5	4	4	P0	YES
Restricted zone breach	5	5	4	P0	YES
Kiosk audio/visual alert	5	5	5	P0	YES
Incident timeline/evidence	5	5	5	P0	YES
Supervisor mobile alert + evidence clip	5	5	5	P0	YES
Risk score / severity	5	5	5	P0	YES
Heatmap	4	5	5	P1	YES
Fall/person-down	5	3	3	P1	IF FOOTAGE
Safety vest	4	5	4	P1	IF MODEL
Fire/smoke	5	3	3	P1	IF FOOTAGE
Work-at-height/harness	5	3	3	P1	IF FOOTAGE
Crane swing zone	5	3	2	P2	FUTURE
Housekeeping/debris	3	3	3	P2	FUTURE
PPE gloves/eye protection	3	3	2	P2	FUTURE
ANPR	3	3	2	P2	FUTURE
Theft/tampering	3	2	2	P3	FUTURE
Predictive risk forecasting	5	2	1	P3	FUTURE
Full digital twin	4	1	1	P3	FUTURE

MVP target

Minimum winning demo:

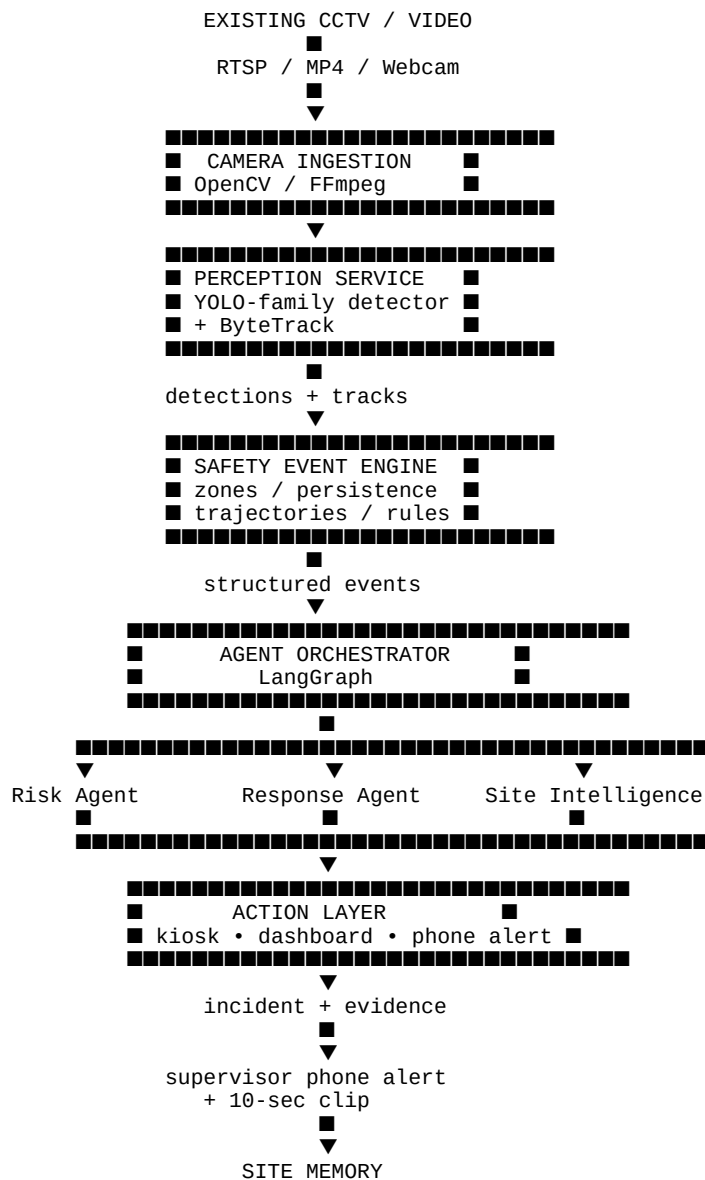
- Person tracking
- PPE violation
- Vehicle detection
- Person–vehicle proximity
- Configurable danger zone
- Local kiosk warning
- Risk prioritization
- Incident evidence
- **Supervisor mobile alert + 10-second evidence clip**
- Heatmap

- Supervisor dashboard

If footage supports it, add **fall detection** or **fire/smoke** as a bonus.

6. Exact MVP architecture

6.1 High-level architecture



6.2 The key architectural rule

YOLO is the eyes, not the brain.

Do not send every video frame to an LLM.

- Computer vision handles fast frame-level perception.
- Tracking creates persistent object identities.

- The event engine turns detections into meaningful events.
- Agents reason over structured events.
- The response layer acts.
- The system verifies whether the risk clears.

This is both technically appropriate and easy to defend.

7. Technology stack

7.1 Recommended hackathon stack

Computer vision

- Python
- OpenCV 4.5+ (Apache 2)
- YOLO-family detector
- ByteTrack
- FFmpeg for video/RTSP handling

Recommended path: use a construction/PPE-trained YOLO-compatible model rather than spending most of the hackathon training from scratch.

Useful public datasets include:

- Construction Site Safety dataset: 717 images, 25 classes including person, hardhat, no-hardhat, safety vest, no-safety vest, machinery, vehicles and excavator; CC BY 4.0.
- Hard Hat Workers dataset: ~7,041 images; public domain.
- SHWD: 7,581 images with 9,044 helmet-positive objects and 111,514 normal-head objects.

Important commercial licensing note

Ultralytics currently states that its YOLO software/models are available under AGPL-3.0 for open-source use, while proprietary/commercial embedding requires an applicable commercial license. Therefore:

- **Hackathon prototype:** use it if event rules permit and document the dependency.
- **Commercial product:** either obtain the appropriate license or evaluate an Apache-2.0 detector such as YOLOX/RT-DETR and validate model suitability.

YOLOX is Apache-2.0; RT-DETR's official repository is Apache-2.0.

Backend

- FastAPI
- Pydantic
- Python asyncio
- WebSockets for live dashboard events

Event transport

For the 30-hour MVP:

- FastAPI internal event bus / asyncio queue

For production:

- Redis Streams or NATS
- Kafka only at much larger scale

Agent orchestration

- LangGraph
- Pydantic structured outputs
- Optional LLM for:
 - incident explanation
 - report generation
 - natural-language site queries
 - policy interpretation
 - post-event reasoning

Do **not** use an LLM for raw frame processing.

MCP

Use MCP only where it adds real interoperability:

- get_camera_status
- get_active_incidents
- get_zone_rules
- get_recent_events
- create_incident
- acknowledge_alert
- get_site_risk_summary

MCP is an open protocol with an MIT-licensed specification repository.

Database

Hackathon:

- SQLite + SQLAlchemy

Production:

- PostgreSQL
- pgvector if semantic incident search becomes useful
- object storage for evidence clips
- short incident evidence clips (target: 10 seconds, ideally 5 seconds before + 5 seconds after trigger)

Frontend

- React

- Vite
- Tailwind CSS
- WebSocket client
- Recharts
- Leaflet only if using a real map; for construction sites, prefer a **site-plan/SVG map** over a geographic map.

Kiosk

Recommended:

- React kiosk UI
- Chromium in fullscreen/kiosk mode on Ubuntu
- WebSocket connection to FastAPI
- Local audio through browser/OS
- Optional physical buzzer/LED later

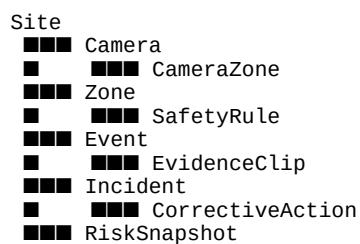
This is much faster than building a native kiosk application.

Deployment

- Docker Compose
- NVIDIA Container Toolkit if a local NVIDIA GPU is available
- Nginx reverse proxy in production
- Systemd service for auto-start on Ubuntu
- Local network first; cloud is optional

8. Data model

8.1 Core entities



8.2 Suggested database schema

sites

- id
- name
- location
- timezone
- configuration_json

- created_at

cameras

- id
- site_id
- name
- stream_uri
- source_type
- fps
- resolution
- status
- created_at

zones

- id
- site_id
- name
- polygon_json
- risk_level
- allowed_classes
- active_hours

safety_rules

- id
- zone_id
- rule_type
- threshold_json
- severity
- enabled

detections

- id
- camera_id
- track_id
- class_name
- confidence
- bbox_json
- timestamp

events

- id
- site_id
- camera_id

- zone_id
- event_type
- severity
- confidence
- started_at
- ended_at
- status
- evidence_uri
- context_json

incidents

- id
- event_id
- title
- severity
- state
- assigned_to
- root_cause
- corrective_action
- created_at
- resolved_at
- notification_status
- notified_at
- notification_channel

evidence_clips

- id
- event_id
- camera_id
- start_time
- end_time
- duration_seconds
- storage_uri
- thumbnail_uri
- created_at

Purpose: preserve a short, reviewable evidence window around the detected event without retaining unnecessary continuous footage.

risk_snapshots

- id
- site_id
- zone_id

- timestamp
- risk_score
- event_count
- dominant_risk

9. Event schema

Every safety event should become a structured object.

```
{
  "event_id": "evt_00124",
  "site_id": "SITE_01",
  "camera_id": "CAM_04",
  "zone_id": "LOADING_ZONE",
  "type": "VEHICLE_PEDESTRIAN_PROXIMITY",
  "severity": "CRITICAL",
  "confidence": 0.91,
  "started_at": "2026-09-04T10:32:11",
  "objects": [
    {
      "track_id": "person_17",
      "class": "person"
    },
    {
      "track_id": "truck_03",
      "class": "vehicle"
    }
  ],
  "context": {
    "distance_px": 74,
    "trajectory_intersection": true,
    "persistence_seconds": 1.8
  },
  "recommended_action": "CLEAR_VEHICLE_PATH",
  "evidence_uri": "clips/evt_00124.mp4",
  "evidence_duration_seconds": 10,
  "notification": {
    "channel": "telegram",
    "recipient_role": "zone_supervisor",
    "status": "sent"
  }
}
```

10. Agent architecture

Agent 1 — Vision Agent

Input: camera frame stream

Output: detections/tracks

Responsibilities:

- Run detector
- Track objects
- Normalize detections
- Attach camera/zone metadata

Agent 2 — Safety/Event Agent

Input: detections/tracks + zone rules

Output: safety events

Examples:

- Person without helmet
- Person inside restricted zone
- Vehicle entering pedestrian zone
- Person/vehicle proximity
- Person down

Agent 3 — Risk Agent

Input: events + site context

Output: severity + priority

Suggested first-pass score:

$$\text{Risk} = \text{Severity} \times \text{Exposure} \times \text{Persistence} \times \text{ProximityFactor}$$

Example:

- Missing helmet: medium
- Restricted excavation entry: high
- Vehicle/pedestrian collision trajectory: critical
- Suspected worker down: critical

Agent 4 — Response Agent

Input: prioritized event

Output: actions

Examples:

- kiosk warning
- visual alert
- supervisor notification with incident summary
- 10-second evidence clip attachment/link
- incident creation
- escalation if the event persists or remains unacknowledged
- repeated warning
- escalation

Agent 5 — Site Intelligence Agent

Input: historical events/incidents

Output:

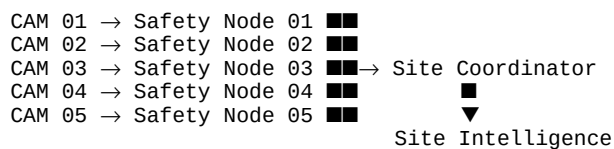
- risk heatmap
- recurring hotspot
- shift trend
- event summaries
- suggested corrective actions

11. Swarm model

The swarm is not “five LLMs talking to each other.”

A more credible definition is:

A distributed set of safety nodes that independently perceive and reason about local zones while exchanging structured events with a site-level coordination layer.



Why this matters:

- lower latency
- local resilience
- reduced bandwidth
- modular deployment
- fault isolation
- camera/zone-specific rules
- cross-camera correlation

12. Critical use cases

Use case A — Missing helmet

- Camera sees person.
- Vision model detects person/head/helmet state.
- Event engine confirms violation persists.
- Risk agent assigns medium severity.
- Response agent triggers nearby kiosk.
- Kiosk says: **“Safety warning. Helmet required.”**

- If violation clears, event closes.
- If it persists, supervisor receives escalation.
- Incident/event is logged.

Use case B — Vehicle/pedestrian conflict

- Vehicle enters loading zone.
- Person is tracked in same zone.
- Trajectories are calculated.
- Distance/proximity threshold is crossed.
- Risk agent marks event high/critical.
- Local kiosk warns the area.
- Supervisor receives alert.
- Event evidence is stored.
- Site heatmap updates.

Use case C — Restricted zone

- Define polygon around excavation/crane/loading zone.
- Person enters polygon.
- Persistence threshold prevents one-frame false alarms.
- Event is generated.
- Kiosk warns locally.
- Escalation happens if person remains.

Use case D — Fall/person down

If footage supports it:

- track person posture/position
- detect rapid change
- require persistence
- create suspected fall event
- critical local alert
- supervisor escalation
- evidence clip

Important: call this “suspected fall/person-down” rather than claiming perfect medical-grade fall detection.

13. Kiosk design

The kiosk is the physical interface between AI and the site.

Normal state

- Site status
- Camera count
- Active workers/vehicles
- Risk map
- Current alerts

Warning state

- Large hazard type
- Zone
- Camera
- Risk
- Clear corrective instruction

Example:

[illegible]

14. Supervisor Mobile Incident Response

The system should not stop at detecting a hazard or updating a dashboard. For actionable incidents, the **Response Agent** sends the responsible supervisor a concise alert on their phone.

14.1 What the supervisor receives

Example:

■ HIGH-RISK SAFETY INCIDENT

Vehicle-Person Proximity

- Block A – Loading Zone
- Camera 04
- 11:32:14 AM
- Confidence: 91%

Action: Clear vehicle path

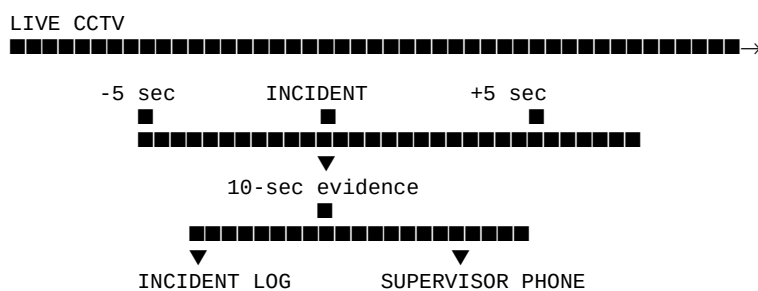
■ 10-second evidence clip attached

The message should contain:

- incident type
- severity
- site / zone
- camera
- timestamp
- confidence or risk score
- recommended action
- short evidence clip or secure clip link

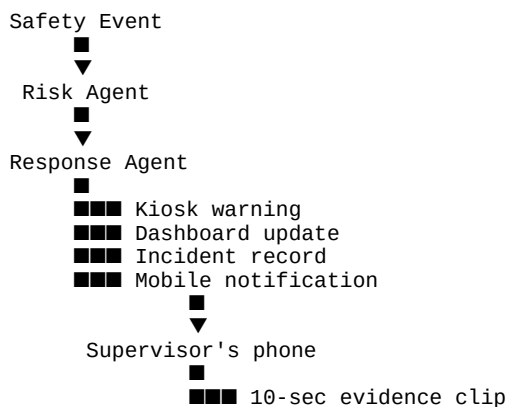
14.2 Evidence clip design

Use a rolling video buffer so the system can capture context immediately around the trigger:



Preferred MVP behavior: keep a rolling buffer, preserve the previous ~5 seconds, then record ~5 seconds after the trigger. If post-trigger capture adds too much latency or complexity, ship a reliable pre-buffered clip first and extend it later.

14.3 Notification architecture



For the 30-hour build, use the fastest reliable channel available to the team (for example, Telegram or an already-configured messaging/webhook integration). Keep the notification provider replaceable so production deployments can use SMS, WhatsApp, email, app push, or an EHS integration as appropriate.

14.4 Alert policy

Do not notify the supervisor for every frame-level detection. The event engine should debounce and group repeated detections.

- Informational → kiosk/dashboard only
- Warning → kiosk + dashboard

- High → kiosk + supervisor phone
- Critical → kiosk/alarm + supervisor phone + incident record

Apply cooldowns and deduplication to prevent alert fatigue.

14.5 Why this matters to the product story

The complete intervention loop becomes:

Detect → Understand → Warn → Escalate → Evidence → Resolve → Learn

The important product outcome is not merely that AI saw the hazard. It is that the **right person receives enough context to act immediately**.

14.6 Competitive honesty

This feature should not be presented as uniquely novel. Current construction-safety platforms such as Xorth already advertise alerts routed to responsible personnel with timestamped evidence clips, and Protex provides real-time response workflows. The defensible Synapz hypothesis is the **integration of mobile escalation with distributed edge nodes, local kiosk intervention, event-level agent reasoning, site-specific context and a unified incident/evidence loop**. This is a product differentiation hypothesis to validate, not a claim of exclusive capability.

15. CCTV / video hunting plan

Priority 1 — Find footage that shows multiple useful classes

Search YouTube / public datasets for:

- “construction site CCTV workers hard hat”
- “construction site surveillance camera workers”
- “construction site CCTV vehicle workers”
- “construction site safety helmet detection video”
- “construction worker CCTV excavator”
- “construction site forklift pedestrian”
- “construction site fall detection CCTV”
- “construction site PPE violation CCTV”
- “construction site safety monitoring camera”
- “construction site timelapse workers safety”

Priority 2 — Public datasets

Construction Site Safety

Roboflow Universe dataset:

- 717 images

- 25 classes
- person, hardhat, no-hardhat, safety vest, no-safety vest, vehicles, excavator, machinery, etc.
- CC BY 4.0

Hard Hat Workers

Public-domain dataset:

- ~7,041 images
- person/head/helmet labels

SHWD

- 7,581 images
- 9,044 helmet-positive objects
- 111,514 normal-head objects

Strategy

Use **real video for temporal demonstration** and public datasets for model evaluation/fine-tuning.

Do not pretend an image dataset validates temporal safety behavior.

16. Footage selection checklist

Before downloading a video, inspect:

Criterion	Ideal
Camera angle	Fixed CCTV-like
Resolution	≥720p
People visible	Yes
PPE visible	Clear
Vehicles visible	At least one
Movement	Continuous
Camera shake	Low
Occlusion	Moderate
Lighting	Daylight first
Scene duration	≥30–60 sec
License	Reusable / permitted

The best demo footage is not necessarily the most dramatic. It is footage where **the model can reliably see the event**.

17. Validation experiment design

SCT rewards evidence. We should record measurements during the event.

Experiment 1 — Detection quality

Create a small manually reviewed test set from the actual demo footage.

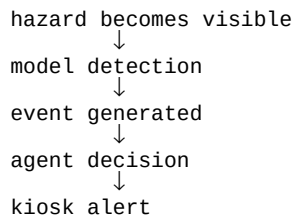
For each scenario measure:

- true positives
- false positives
- false negatives
- precision
- recall

Do not report generic model benchmark scores as if they represent your site.

Experiment 2 — End-to-end latency

Measure:



Record:

- average latency
- p95 latency if possible

Target for the demo: **sub-2-second local alert where hardware permits**, but report the measured value.

Experiment 3 — Alert fatigue

Compare:

- alert every frame

vs.

- persistence + cooldown + event deduplication

Metric:

alerts generated per minute of continuous violation

The product should generate **one meaningful event**, not 30 alerts.

Experiment 4 — Multi-camera correlation

If two clips/cameras are available:

- identify vehicle at entrance

- correlate with person in internal zone
- demonstrate site-level context

Experiment 5 — Human usefulness

Have one team member act as a supervisor.

Give them:

- raw CCTV
- AI event stream

Ask:

- Which hazard would you respond to first?
- Could you identify the location quickly?
- Is the alert understandable?
- Is the evidence sufficient?

This is a lightweight usability experiment.

18. Validation log template

At each checkpoint answer:

What did we believe?

“Continuous CCTV AI can detect selected construction hazards with sufficiently low latency to trigger local action.”

What did we test?

“Helmet/PPE and vehicle/person proximity on our chosen footage.”

What happened?

Record actual precision, false positives, latency and failure cases.

What did we learn?

Example:

“Helmet detection was strong, but distant workers caused missed detections.”

What did we decide?

Example:

“We changed camera 2’s confidence threshold and made helmet alerts require persistence.”

This directly matches the Startathon’s evidence-based pivot philosophy.

19. Business model

18.1 Customer

Primary:

- Construction companies
- EPC contractors
- Infrastructure developers
- Large project contractors
- Industrial construction operators

Secondary:

- Safety/EHS consultants
- Facility/security integrators
- CCTV installation companies
- Insurance/risk partners

18.2 Buyer

Likely economic buyer:

- Project director
- Head of EHS/safety
- Operations director
- Infrastructure program owner

Operational users:

- Safety officer
- Site supervisor
- Control-room operator

18.3 Pricing model

Do not claim this is validated pricing yet.

Suggested commercial model

Pilot

- 30-day single-zone/single-site pilot
- fixed deployment fee
- measured baseline vs AI-assisted safety workflow

Site subscription

Illustrative:

- ■2–5 lakh/site/year software + support
- deployment/hardware charged separately

Enterprise

- volume pricing across multiple sites
- integrations
- custom models
- centralized portfolio analytics

Better than per-user pricing

Safety monitoring is site infrastructure. Price by:

- site
- camera count
- enabled modules
- required edge hardware
- support tier

20. Unit economics model

The first business spreadsheet should use:

```

Annual contract value
- edge hardware amortization
- installation
- cloud/storage
- support
- model inference cost
- connectivity
= contribution margin

```

Example scenario only:

100 sites × ■3 lakh/year = ■3 crore ARR

This is an illustration, **not TAM**.

The first real business experiment should determine:

- cameras/site
- GPU requirement
- installation hours
- support cost
- customer willingness to pay
- acceptable false-alert rate

21. TAM / SAM / SOM methodology

Avoid inventing a giant “AI construction safety market” number.

TAM

Use the number of addressable construction sites/projects that could reasonably use CCTV-based safety intelligence × annual contract value.

Possible top-down context:

- India's construction sector is large and construction accounts for about 12% of employment in the latest PLFS 2025 broad-industry data.
- Global AI-in-construction market estimates are growing rapidly, but third-party market reports should be treated as directional rather than as our actual obtainable market.

SAM

Define:

- India
- medium/large active construction projects
- sites with existing CCTV/IP cameras
- safety teams with a budget for monitoring technology

SOM

Bottom-up:

Targetable sites in first 3 states
× reachable sites/year
× expected annual contract value

Example:

100 reachable sites
× ■3 lakh/site/year
= ■3 crore potential annual revenue

The number of reachable sites must be validated through customer discovery.

22. ROI model

Do not promise “40% fewer accidents” without a controlled customer study.

Instead, calculate:

Direct savings

- reduced routine monitoring burden
- reduced manual incident documentation
- reduced time to detect/respond

- fewer project interruptions
- lower investigation time

Risk value

- near misses caught earlier
- hazards corrected before incidents
- recurring hotspots identified

Operational value

- equipment/vehicle flow visibility
- zone occupancy
- safety trend reporting

The customer ROI calculation should be:

$$\text{ROI} = \frac{(\text{avoided cost} + \text{productivity savings} + \text{compliance/admin savings} - \text{annual platform cost})}{\text{annual platform cost}}$$

23. Market evidence

India's PLFS 2025 annual report reports construction at about **12.0% of workers' usual-status employment** in the relevant broad-industry distribution. [Source: OSHA Construction Safety and Health Program — <https://www.osha.gov/etools/construction/safety-health-program>]

Invest India materials have historically highlighted construction as one of India's largest employment and investment sectors. Current IBEF material reports substantial cumulative FDI into construction development and infrastructure construction activity through March 2026. [Source: OSHA Construction Safety and Health Program — <https://www.osha.gov/etools/construction/safety-health-program>]

Grand View Research estimates the global AI-in-construction market at **\$5.2B in 2025**, with a projected \$6.6B in 2026 and \$35B by 2033. Treat this as external market research, not as a directly addressable Team Synapz market figure. [Source: OSHA Construction Safety and Health Program — <https://www.osha.gov/etools/construction/safety-health-program>]

24. Risks and limitations

Camera coverage

A camera cannot detect what it cannot see.

Mitigation: camera-to-risk coverage mapping.

False positives

Repeated incorrect alerts cause people to ignore the system.

Mitigation: persistence, confidence thresholds, cooldowns and event deduplication.

False negatives

A missed hazard can be dangerous.

Mitigation: human oversight, conservative high-risk rules and never positioning AI as the sole safety control.

Lighting/weather

Night, dust, rain and glare reduce vision performance.

Mitigation: site-specific validation and additional sensors/cameras where necessary.

Network outage

Cloud-only systems may fail.

Mitigation: local edge inference and local kiosk operation.

Privacy

CCTV contains people.

Mitigation:

- process locally where possible
- avoid facial recognition
- blur faces in stored evidence if appropriate
- RBAC
- retention limits
- audit logs
- transparent worker communication

Regulatory responsibility

The platform is an assistive safety layer, not a statutory replacement for safety officers, inspections or required controls.

25. Judge Q&A

“Protex already does this. Why do you exist?”

Answer:

“Protex validates the market, and we respect that overlap. We aren't claiming the invention of AI safety monitoring. Our hypothesis is that construction needs a more site-native safety layer: distributed local safety nodes, physical worker-facing intervention, cross-camera event context and an India-first deployment/economic model. The 30-hour MVP is designed to test

that workflow rather than claim a finished moat.”

“Why YOLO?”

“YOLO gives us fast visual perception. We need object-level detections and tracking in real time. The agents then reason over structured events rather than raw frames.”

“Why agents?”

“We don’t need agents to detect a helmet. We need orchestration when multiple signals interact—for example, a vehicle, a person, a danger zone and persistence can combine into one high-risk event. Agents are used for event reasoning, prioritization, response and site intelligence.”

“Why swarm?”

“Construction risk is spatially distributed. Local camera nodes can make fast decisions near their zones while a site-level coordinator correlates events across cameras. We exchange structured events instead of streaming every frame to a central intelligence layer.”

“Why not just have a CCTV operator?”

“An operator remains valuable, but human attention is limited. We are turning the cameras into an always-on first layer that surfaces only actionable events and gives the operator context.”

“Are you replacing safety officers?”

“No. We are augmenting them. Safety officers remain responsible for policy, controls, investigation and judgment. Our system increases their coverage and response speed.”

“What if the AI is wrong?”

“We never treat one detection as ground truth for a consequential decision. We use confidence, persistence and contextual rules, and escalate appropriately. High-impact actions remain human-controlled.”

“Why a kiosk?”

“A remote alert may reach a supervisor after the event. A local kiosk can warn people already inside the risk zone. It is the physical actuator of the intelligence layer.”

“Why not just send an app notification?”

“That’s useful for supervisors, but the person at risk may not be looking at a phone. Local audio/visual intervention is closer to the hazard.”

“How will you make money?”

“B2B site-level subscription plus deployment and edge hardware. We would validate pricing through paid pilots. We are not pretending our initial price is validated yet.”

“How will you get customers?”

“Start with one-site pilots through construction companies, EPC firms, CCTV integrators and EHS partners. The existing-camera retrofit makes the pilot less disruptive.”

“What is your moat?”

“Today, the moat is a hypothesis, not a claim. Over time it becomes site-specific safety data, calibrated models, event graphs, deployment expertise, integrations and workflow lock-in.”

“What if internet goes down?”

“The safety-critical perception and local alert path is designed to run on-site. Cloud services are for aggregation, reporting and portfolio intelligence.”

“Why not use sensors instead of cameras?”

“Where sensors are superior, we should use them. Cameras are valuable because many sites already have them and they provide rich contextual information without requiring workers or vehicles to wear new devices.”

“Can CCTV really detect a fall?”

“Not perfectly. Our MVP treats it as suspected fall/person-down detection and validates it on the actual camera angle. We would not market it as a guaranteed rescue system without site validation.”

“How do you prevent alert fatigue?”

“Event persistence, deduplication, cooldowns, severity prioritization and escalation. We will actually measure alerts per minute in the hackathon.”

“What did you learn during the hackathon?”

The answer must be real. Never invent it. Use:

“We believed X. We tested Y. We observed Z. Therefore we changed A.”

26. 30-hour task board

Team member A — Vision / Edge

Hours 0–3

- Inspect footage
- Choose 2–3 scenarios
- Set up detector
- Verify GPU

Hours 3–8

- Video ingestion
- YOLO inference
- tracking
- annotated video

Hours 8–14

- PPE logic
- vehicle/person detection
- zone geometry

Hours 14–19

- proximity/trajectory logic
- latency measurements
- false-positive tuning

Hours 19–24

- optimize inference
- create evidence clips
- stabilize demo

Hours 24–30

- test all scenarios
- benchmark
- support final demo

Team member B — Agents / Backend

Hours 0–4

- event schema
- database schema
- API contract

Hours 4–9

- event engine
- rule engine
- persistence/cooldowns

Hours 9–15

- LangGraph orchestration
- Risk Agent
- Response Agent

Hours 15–20

- Site Intelligence Agent
- incident logging
- heatmap aggregation

Hours 20–25

- WebSocket alerts
- failure handling
- validation metrics

Hours 25–30

- end-to-end integration
- demo orchestration
- judge Q&A preparation

Team member C — Product / Kiosk / Business

Hours 0–4

- user journey
- kiosk wireframe
- site map
- demo story

Hours 4–10

- React kiosk
- dashboard
- alert states

Hours 10–16

- heatmap
- incident timeline
- camera cards

Hours 16–21

- UX polish
- kiosk fullscreen
- audio

Hours 21–25

- validation dashboard
- evidence screenshots
- business model

Hours 25–30

- presentation
- demo script
- judge Q&A
- final rehearsal

27. Integration checkpoints

Checkpoint 1 — Hour 5

Must have:

- footage
- detector
- basic tracking
- agreed MVP
- baseline assumptions

Checkpoint 2 — Hour 12

Must have:

- one end-to-end detection
- structured event
- backend persistence

Checkpoint 3 — Hour 18

Must have:

- risk decision
- kiosk response
- incident creation

Checkpoint 4 — Hour 24

Must have:

- 3 scenarios
- heatmap
- metrics

- stable demo

Checkpoint 5 — Hour 27

Feature freeze.

No new major feature after this point.

Hours 27–30

- bug fixing
- evidence capture
- presentation
- rehearsal

28. Repository structure

```
synapz-safety/
├── apps/
│   ├── api/
│   ├── kiosk/
│   ├── dashboard/
│   └── vision/
│       ├── detector/
│       ├── tracker/
│       ├── preprocessing/
│       └── inference.py
│   ├── safety/
│   │   ├── zones/
│   │   ├── rules/
│   │   ├── events/
│   │   └── risk/
│   ├── agents/
│   │   ├── graph.py
│   │   ├── risk_agent.py
│   │   ├── response_agent.py
│   │   └── intelligence_agent.py
│   ├── data/
│   │   ├── videos/
│   │   ├── evidence/
│   │   └── configs/
│   ├── db/
│   ├── models.py
│   └── migrations/
├── tests/
├── docker-compose.yml
└── README.md
```

29. API contract

GET /api/cameras

Returns camera status.

GET /api/events?status=active

Returns active safety events.

POST /api/events

Creates structured event.

POST /api/events/{id}/ack

Acknowledges an event.

POST /api/incidents

Creates incident.

GET /api/incidents

Incident history.

GET /api/risk-map

Returns risk values by zone.

WebSocket

/ws/site/{site_id}

Push:

- new event
- event escalation
- event resolved
- kiosk instruction
- camera status

30. Product roadmap

Phase 0 — Hackathon

- 3–4 safety scenarios
- existing video
- local kiosk
- dashboard
- risk engine
- event memory

Phase 1 — Pilot

- 1 real construction site
- 5–20 cameras
- on-site GPU
- site calibration

- measured baseline
- worker feedback

Phase 2 — Production

- multi-site management
- EHS integrations
- multilingual alerts
- stronger event graph
- edge fleet management
- automated reports

Phase 3 — Autonomous safety intelligence

- predictive hotspots
- site-specific model adaptation
- digital site graph
- sensor fusion
- policy-aware agent workflows
- portfolio-level benchmarking

31. Final pitch architecture

The final pitch should evolve from the baseline deck into:

Slide 1 — The problem

“Construction sites already have eyes. Why don't those eyes act?”

Slide 2 — Current workflow

Show:

CCTV → human observation → communication → intervention

Highlight latency and attention bottleneck.

Slide 3 — Evidence

Construction safety + actual customer/user discovery + your experiment evidence.

Slide 4 — Solution

Sahayi-like positioning is not appropriate here; use the actual Team Synapz product name.

Recommended product line:

“Autonomous Construction Safety Intelligence.”

Slide 5 — Live system

CCTV → YOLO → Event Engine → Agents → Kiosk.

Slide 6 — Demo

Helmet violation → local warning.

Slide 7 — Demo

Vehicle/person conflict → critical alert.

Slide 8 — Site intelligence

Heatmap + incidents + recurring hotspots.

Slide 9 — Why this architecture

Explain:

- edge
- YOLO
- event engine
- agents
- local action
- human oversight

Slide 10 — Competition

Acknowledge competitors honestly.

Slide 11 — Differentiation

Construction-first + distributed + local intervention + India-first.

Slide 12 — Business

Pilot → site subscription → enterprise.

Slide 13 — Validation

Actual metrics:

- precision
- recall
- latency
- alert reduction
- usability observation

Slide 14 — Roadmap

30-hour MVP → pilot → production.

Slide 15 — Closing

“We are not replacing the safety officer. We are giving every safety officer an AI that never looks away.”

32. Demo choreography

The demo should be scripted so judges see a complete closed loop.

Scene 1

Play footage.

Narrate:

“This is ordinary CCTV. Right now, it records what happens. Our system is about to interpret it.”

Scene 2

Worker without helmet.

System:

- bounding box
- PPE event
- risk score
- kiosk warning

Say:

“The system didn't just detect a person. It converted a visual observation into a safety event and delivered the corrective action locally.”

Scene 3

Vehicle/pedestrian interaction.

System:

- track IDs
- trajectory
- proximity
- critical alert
- kiosk

Say:

“This is where context matters. Neither object is inherently dangerous. Their interaction is.”

Scene 4

Dashboard

Show:

- active incident
- evidence clip
- heatmap
- camera
- timestamp

Scene 5

Resolution

Worker/vehicle leaves risk area.

System:

- event resolved
- incident state updated

Say:

“The system verifies whether the risk actually cleared.”

33. What NOT to claim

Never say:

- “100% accurate”
- “prevents all accidents”
- “replaces safety officers”
- “eliminates the need for supervisors”
- “our technology is completely unique”
- “our AI understands everything”
- “swarm intelligence guarantees safety”
- “we can detect every construction hazard”
- “our model works on every camera”

Instead say:

- “additional safety layer”
- “assists safety teams”
- “measured on our test footage”
- “site-specific calibration”
- “human oversight”
- “our current MVP supports...”

34. Immediate build decisions

Build now

- YOLO-compatible detector
- ByteTrack
- OpenCV/FFmpeg
- FastAPI
- SQLite
- React/Vite kiosk
- WebSockets
- LangGraph
- 5 agents
- rule engine
- heatmap
- incident evidence

Avoid unless everything else is stable

- Kafka
- Kubernetes
- complex cloud deployment
- custom foundation models
- full digital twin
- face recognition
- biometric identification
- elaborate RAG
- autonomous physical machinery control
- 10+ simultaneous agents

33A. Edge AI Architecture — YOLO + Local LLM

Why YOLO is not the product

A likely judge question is:

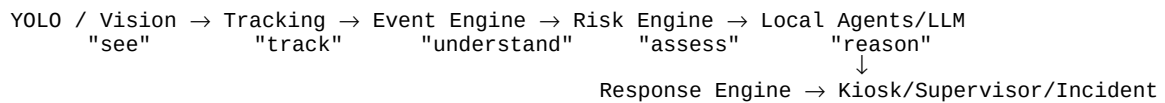
“Isn’t this just a YOLO project?”

The answer should be:

“YOLO is only our perception layer. If this were a YOLO project, we would detect a person without a helmet and draw a bounding box around them. Our product starts where that bounding box ends.”

YOLO answers what objects are visible and where they are. Synapz must answer what safety event is occurring, whether it is dangerous, how long it has persisted, what context is involved, what action should happen, who should be alerted, and what evidence should be recorded.

Product stack



Memorable explanation

“YOLO is the eyes. It isn't the brain, nervous system, or safety officer.”

This distinction should be used whenever judges challenge the technical novelty of the project.

33B. Fully Local / Edge-First Product

Core proposition

Synapz can be designed so that the **safety-critical loop does not require internet connectivity**.

Be precise: cameras communicating with a kiosk over RTSP/Ethernet/Wi-Fi still use a **local network**. The correct claim is:

“Our safety-critical inference and alerting require zero internet connectivity.”

Or:

“The safety loop runs on-site, at the edge. No video needs to leave the construction site.”

The local loop can include:

- YOLO weights stored locally
- tracking
- event/risk logic
- local agents
- local database
- dashboard/kiosk UI
- audio/visual alerts

Cloud connectivity can remain optional for multi-site analytics, remote management, model updates and long-term storage.

33C. YOLO26 on Raspberry Pi

YOLO26 is suitable for edge deployment. Ultralytics documents YOLO26 deployment on Raspberry Pi 4 and 5 and recommends optimized formats such as NCNN for ARM devices. Their Raspberry Pi 5 benchmark reports approximately **67 ms/image for YOLO26n in NCNN** and **126 ms/image in ONNX** at 640 input resolution in their benchmark setup. These are reference benchmarks, not guarantees for Synapz because the complete workload also includes camera decoding, preprocessing, tracking, event logic and UI.

Relevant model sizes

Model	Parameters	Prototype suitability
YOLO26n	~2.4M	Preferred for Raspberry Pi
YOLO26s	~9.5M	Possible, but significantly slower
YOLO26m	~20.4M	Not appropriate for the Pi MVP
YOLO26l/x	Much larger	Not appropriate for the Pi MVP

The first prototype should use **YOLO26n**, exported to an ARM-friendly runtime such as NCNN or ONNX and benchmarked on the exact hardware available.

Product metric

Do not optimize only for YOLO FPS. Measure:

hazard visible → actionable local alert

That is the meaningful end-to-end safety metric.

33D. Local LLM / Llama Agent Architecture

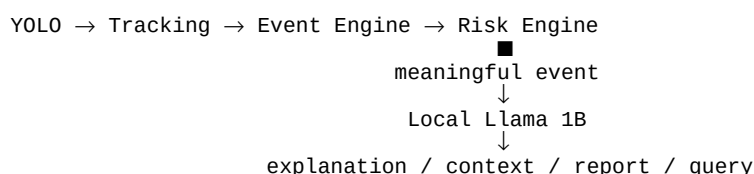
A local language model such as **Llama 3.2 1B** can be used for event-level reasoning, summarization and natural-language interaction.

Meta's Llama 3.2 1B contains approximately **1.23 billion parameters**; the 3B model contains approximately **3.21 billion parameters**. Quantized variants can substantially reduce memory requirements and are the appropriate direction for edge experimentation.

Critical design rule

Do NOT run Llama on every video frame.

Instead:



The LLM receives a **structured safety event**, not raw video.

Example:

```
{
  "event": "vehicle_person_proximity",
  "distance_m": 2.1,
  "duration_sec": 4.8,
  "zone": "Loading_Bay",
  "vehicle": "excavator",
  "severity": "high"
}
```

The LLM can turn that into an explanation or supervisor-facing message.

Safety-critical decision principle

The LLM should not be the sole authority for safety-critical decisions.

Use deterministic rules for the core decision:

```
distance < threshold
AND duration > threshold
AND vehicle moving
AND person in danger zone
↓
HIGH RISK
```

Then use the LLM for:

- incident explanation
- contextual summarization
- supervisor communication
- reports
- natural-language queries
- correlation of already-structured events

Rules decide safety; the LLM provides intelligence around the decision.

33E. Can YOLO26 + Llama Run on the Same Raspberry Pi?

Yes, with the correct workload design.

A Raspberry Pi 5 can run YOLO26n locally, and a quantized 1B-class LLM can be experimented with locally. However, running both continuously at maximum load is **not** the target architecture.

Continuous workload

YOLO26n + Tracker + Event Engine + Risk Rules

Event-driven workload

meaningful event → Local Llama 1B → reasoning/reporting

This makes the system faster, more power-efficient, more deterministic and easier to validate.

Recommended prototype hardware

Raspberry Pi 5 — 8 GB

Suggested stack:

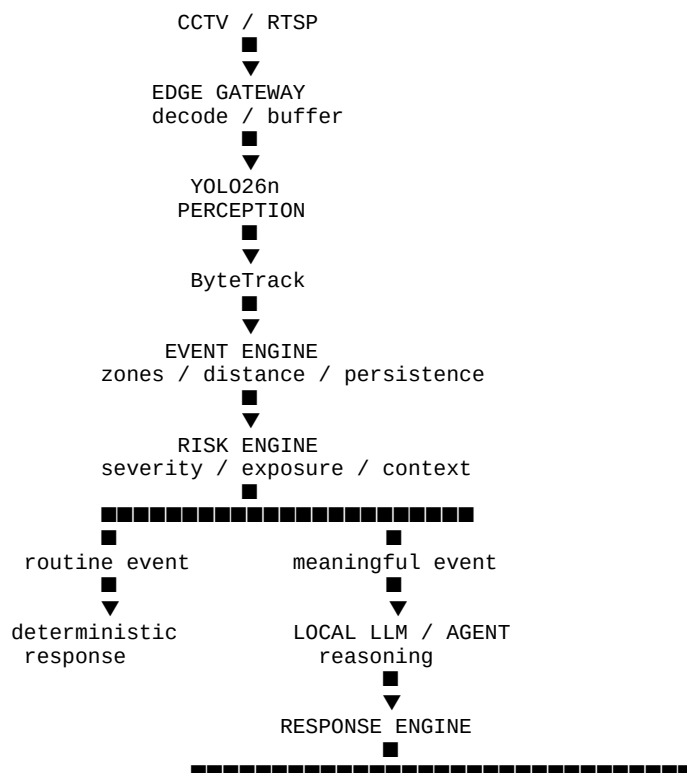
- YOLO26n
- NCNN/ONNX
- ByteTrack
- OpenCV
- FastAPI
- SQLite
- WebSockets
- React/Vite kiosk
- Quantized Llama 3.2 1B
- LangGraph for event-driven agent orchestration

For sustained multi-camera production deployment, a more powerful edge computer or accelerator may be preferable.

Optional acceleration

Ultralytics reports a Raspberry Pi 5 + DEEPX DX-M1 benchmark of approximately **34.6 ms/image for YOLO26n** in its benchmark setup. This is an optional future hardware path, not required for the Startathon MVP.

33F. Recommended Synapz Edge Architecture



▼
KIOSK
AUDIO/UI

▼
SUPERVISOR
DASHBOARD

▼
INCIDENT
EVIDENCE

Architectural principle

Fast perception is continuous. Reasoning is event-driven.

33G. Judge Answer — “Isn't this just a YOLO project?”

Recommended answer:

***“YOLO is actually the easiest part of what we're demonstrating. YOLO tells us that there is a person, a truck or a helmet in an image. It doesn't understand that a worker has been standing inside a danger zone for ten seconds, that a vehicle is approaching that worker, that this is a high-risk event, or what should happen next.

>

Our architecture separates perception from safety intelligence. The vision model generates observations, our event and risk engines turn those observations into contextual safety events, and our response agents determine whether to warn the worker locally, escalate to a supervisor, or create an incident.

>

We also run the safety-critical loop at the edge, so internet connectivity is optional. YOLO is the eyes of Synapz—it is not Synapz.”**

Then immediately demonstrate:

```
YOLO → Vehicle + Person
      → Trajectory / proximity
      → HIGH RISK
      → “CLEAR VEHICLE PATH”
      → Supervisor alert
      → Incident log
```

The demo should prove that the product goes substantially beyond object detection.

33H. Judge Answer — “Why use a local LLM?”

Recommended answer:

“We don't need an LLM to detect a helmet or calculate whether two objects are dangerously close. Those are faster and more reliable with computer vision and deterministic rules. We use a small local LLM only after an event has been structured, where language reasoning is useful—for example, explaining an incident, correlating related events, generating a safety report or answering a supervisor's question. This keeps the real-time safety loop fast and deterministic while allowing the system to remain fully local.”

If asked whether it is running on the kiosk:

“Our architecture supports a quantized 1B-class Llama model locally. The exact deployment will be benchmarked against the chosen Raspberry Pi hardware. If one kiosk is compute-constrained, the LLM can

run on a more capable local site-edge node while perception remains distributed at the kiosks.”

33I. Pitch Terminology

Say

- Edge-first
- Internet-independent safety loop
- Local inference
- Distributed safety nodes
- Event-driven agents
- Structured safety events
- Local intervention
- Site-level safety intelligence
- YOLO as perception
- LLM as event-level reasoning

Avoid

- “Zero network” unless there is genuinely no network.
- “Llama controls safety decisions.”
- “AI replaces safety officers.”
- “YOLO prevents accidents.”
- “Our model is unique.”
- “Nobody else does this.”
- “The LLM watches CCTV.”

Strong positioning

“Synapz is an edge-first construction safety intelligence network. Computer vision continuously sees the site; deterministic safety logic understands physical risk; local agents reason over events; and kiosks intervene where the hazard occurs.”

35. Research sources

Primary / official:

- OSHA Construction Safety and Health Program
- OSHA Recommended Practices for Safety and Health Programs in Construction
- OSHA Incident Investigation
- India BOCW Act / Central Rules
- Ministry of Statistics and Programme Implementation — PLFS 2025

- Ultralytics licensing/legal documentation
- OpenCV licensing
- Model Context Protocol repository
- LangGraph repository

Competitor sources:

- viAct official site and construction solutions
- Protex AI official site, safety, vehicle control, privacy and intelligence pages
- Intenseye official products/safety pages
- Xorth official construction and platform pages
- WorkVis official solutions, PPE, fall detection and privacy pages

Datasets:

- Roboflow Construction Site Safety
- Roboflow Hard Hat Workers
- SHWD Safety Helmet Wearing Dataset

36. Bottom line

Team Synapz should not try to win by having the most AI.

Win by showing:

- We understand the actual safety workflow.
- We know competitors already exist.
- We identified a specific gap.
- We built the smallest system that tests that gap.
- We measured whether it works.
- We changed something because of evidence.
- We can explain every technical decision.
- We have a realistic customer and business model.
- We know what remains uncertain.
- We have a credible path from a 30-hour prototype to a real site pilot.

The strongest one-line description is:

Team Synapz is building an autonomous construction safety intelligence layer that turns existing CCTV into real-time site awareness, local intervention and continuous safety learning.

Appendix A — Direct research links

Competitors

- viAct: <https://www.viact.ai/>

- viAct Area Control: <https://www.viact.ai/solutions/area-control-safety-system>
- Protex AI: <https://www.protex.ai/>
- Protex Vehicle Control: <https://www.protex.ai/safety/vehicle-control>
- Protex Privacy: <https://www.protex.ai/guides/privacy>
- Intenseye: <https://www.intenseye.com/products>
- Intenseye Safety: <https://www.intenseye.com/solutions/safety>
- Xorth Construction: <https://www.xorth.com/constructions>
- Xorth platform: <https://www.xorth.com/>
- WorkVis: <https://www.workvis.io/>
- WorkVis PPE: <https://www.workvis.io/solutions/ppe-compliance/>
- WorkVis Fall Detection: <https://www.workvis.io/solutions/fall-detection/>

Safety / regulatory research

- OSHA Construction Safety & Health Program: <https://www.osha.gov/etools/construction/safety-health-program>
- OSHA Hazard Identification: <https://www.osha.gov/safety-management/hazard-identification>
- OSHA Incident Investigation: <https://www.osha.gov/incident-investigation>
- OSHA Construction Recommended Practices: <https://www.osha.gov/sites/default/files/publications/OSHA3886.pdf>
- India BOCW Act: https://megbocwwb.gov.in/acts/BOCW_Act.pdf
- PLFS 2025: <https://mospi.gov.in/>

Technology

- Ultralytics license: <https://www.ultralytics.com/license>
- Ultralytics pricing/licensing: <https://www.ultralytics.com/pricing>
- YOLOX: <https://github.com/Megvii-BaseDetection/YOLOX>
- RT-DETR: <https://github.com/lyuwenyu/RT-DETR>
- ByteTrack: <https://github.com/FoundationVision/ByteTrack>
- OpenCV license: <https://opencv.org/license/>
- LangGraph: <https://github.com/langchain-ai/langgraph>
- MCP: <https://github.com/modelcontextprotocol/modelcontextprotocol>

Datasets

- Construction Site Safety: <https://universe.roboflow.com/roboflow-universe-projects/construction-site-safety>
- Hard Hat Workers: <https://public.roboflow.com/object-detection/hard-hat-workers>
- SHWD: <https://github.com/njvisionpower/Safety-Helmet-Wearing-Dataset>

New edge-AI sources

- Ultralytics — YOLO26 on Raspberry Pi: <https://docs.ultralytics.com/guides/raspberry-pi>
- Ultralytics — YOLO26 model documentation: <https://docs.ultralytics.com/models/yolo26>
- Ultralytics — Raspberry Pi + DEEPX acceleration benchmark: <https://docs.ultralytics.com/integrations/deepx>
- Meta Llama 3.2 1B model card: <https://huggingface.co/meta-llama/Llama-3.2-1B>